

GENE EXPRESSION DATA ANALYSIS IN BRCA

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BREAST INVASIVE CARCINOMA: OVERVIEW

Invasive breast cancers lack structured architecture, usually showing casual cell infiltration into the tisssues.
It can be categorize mainly into ductal and lobular histologic types:

1 **Invasive Ductal Cancer**

- Grows as a cohesive mass.
- Appears as discrete abnormalities on mammograms.
- Often palpable as a discrete lump, usually smaller than lobular cancers.
- Most common form, accounting for 50-70% of invasive breast cancers.

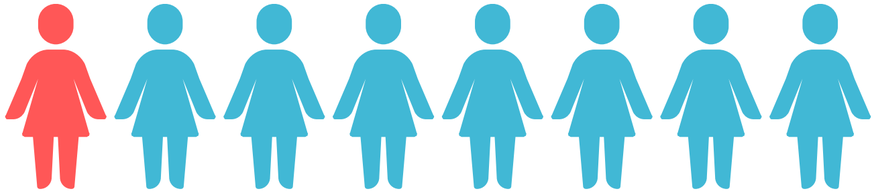
2 **Invasive Lobular Cancer:**

- Tends to permeate the breast in a single-file manner.
- Often remains clinically undetected until extensive, due to its nature of spreading.
- Accounts for 10% of breast cancers.
- Mixed Ductal and Lobular Cancers: Increasingly recognized and described.

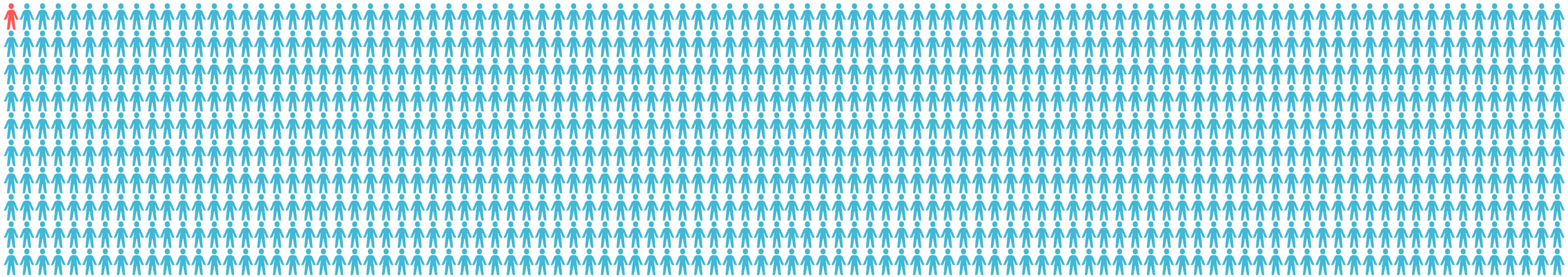
Alkabban FM, Ferguson T. Breast Cancer. 2022 Sep 26. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan–. PMID: 29493913.

SOME STATISTICS

Invasive breast cancer impacts 1 in 8 women in the U.S. during their lifetime



The incidence in men is roughly 1 in 1000



Alkabban FM, Ferguson T. Breast Cancer. 2022 Sep 26. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan–. PMID: 29493913.

DATASET

The dataset comes from TCGA
and it is a matrix formed by:

- 56602 genes
- 1222 samples

Samples are divided into:

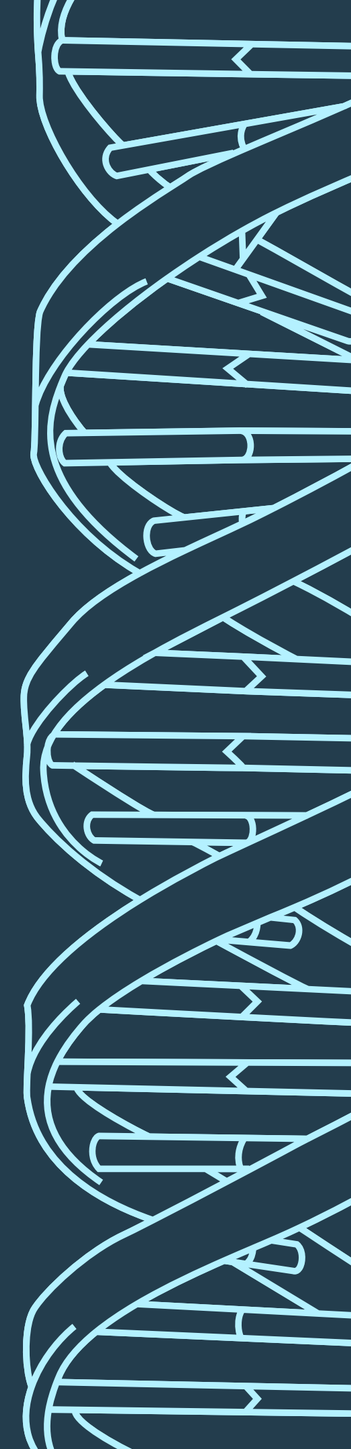
- 113 normal samples from 113 patients
- 1109 cancer samples from 1091 from 1091 patients

In total there are 112 common patients

From clinical data we can extract some interesting informations about patients, where available:

- Mean age at Diagnosis: about 59 years
- Days to Death: Approximately 1584.62 days $\approx$ 4,5 years (for patients who have passed away)

DIFFERENTIALLY EXPRESSED GENES ANALYSIS



PRE-PROCESSING

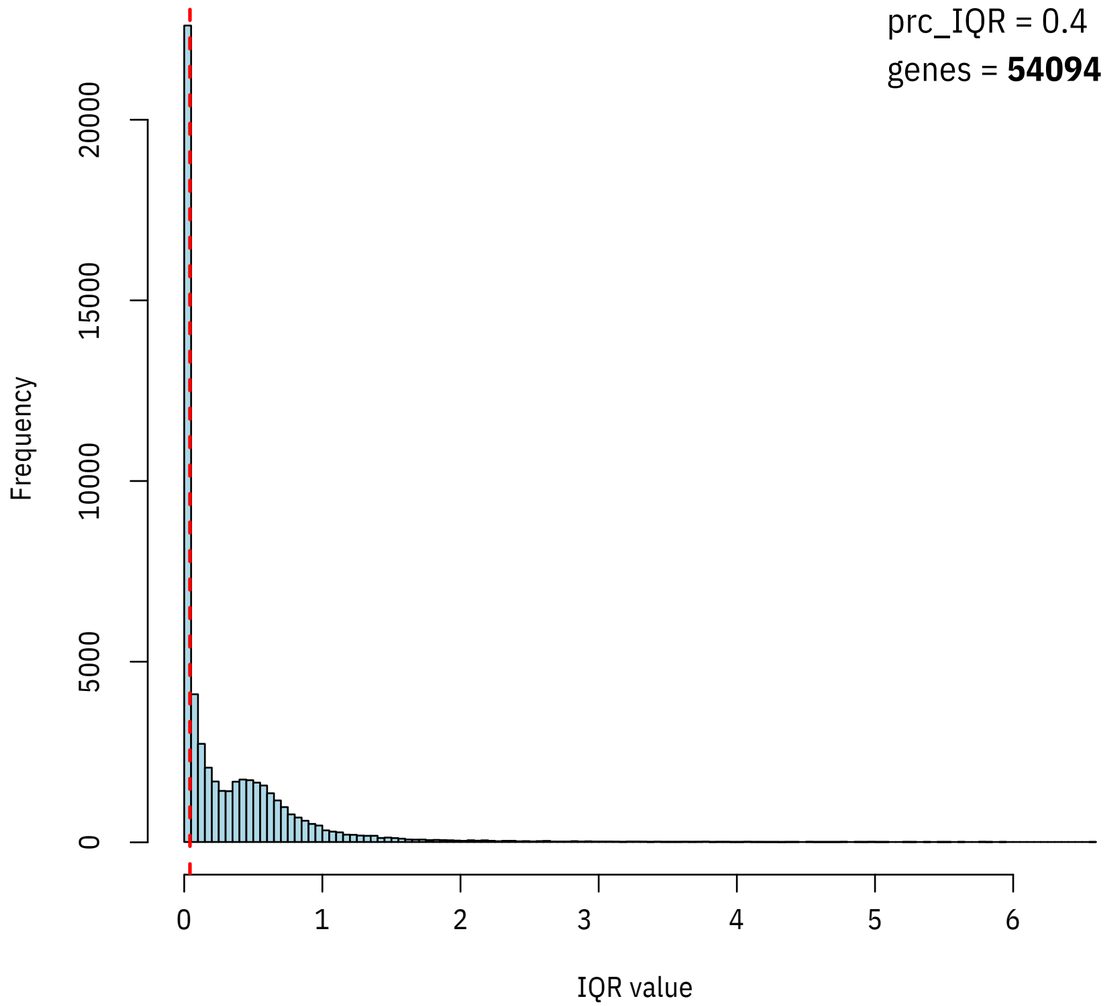
1. Remove genes with mean equal to zero in normal and cancer samples simultaneously.
(54094 genes left)

2. Apply a logarithmic transformation: $\log_2(\text{data} + 1)$

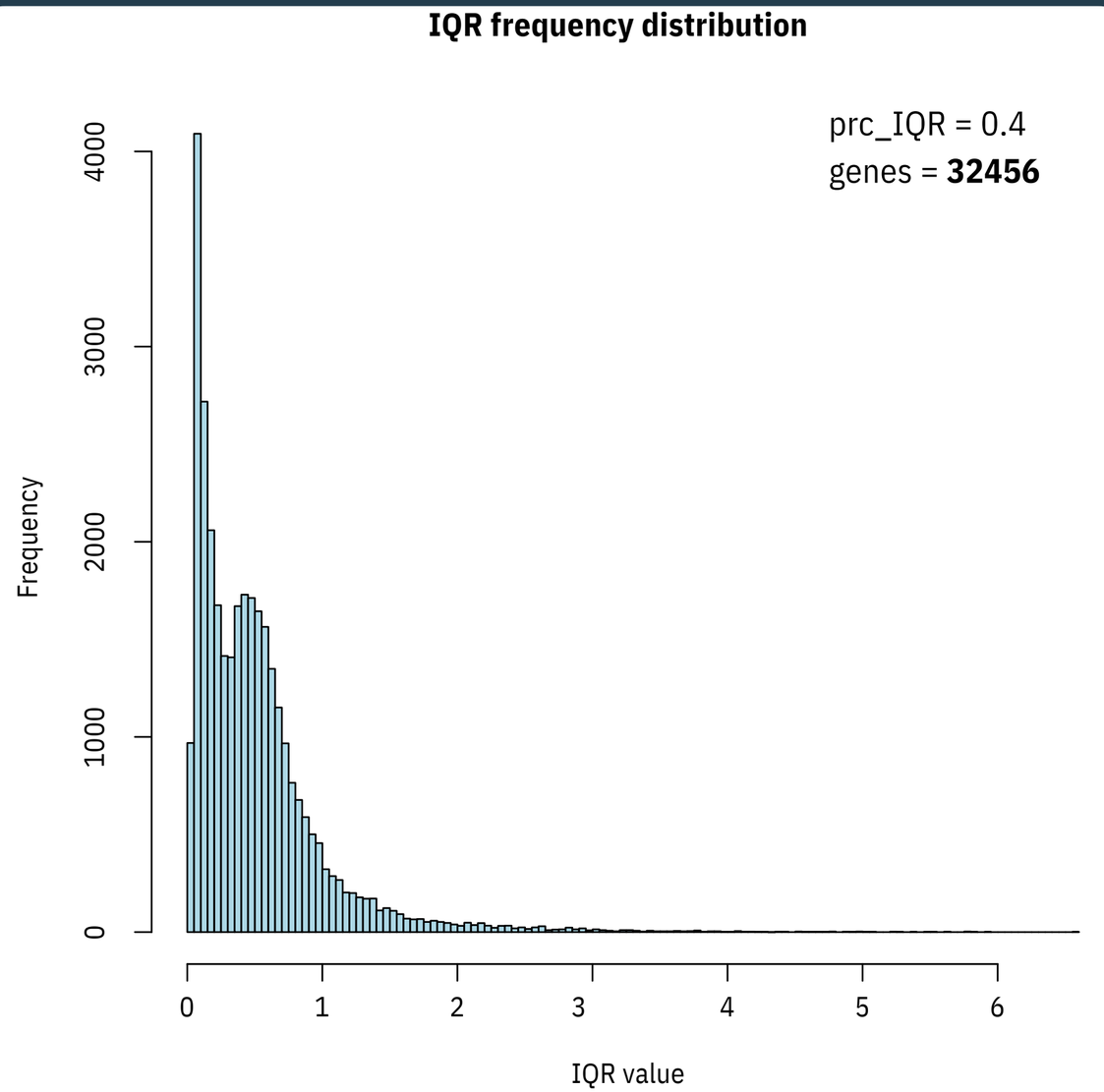
3. Calculate IQR distribution and filter out genes with variation < prc_IQR = 0.4



Before filtering



After filtering

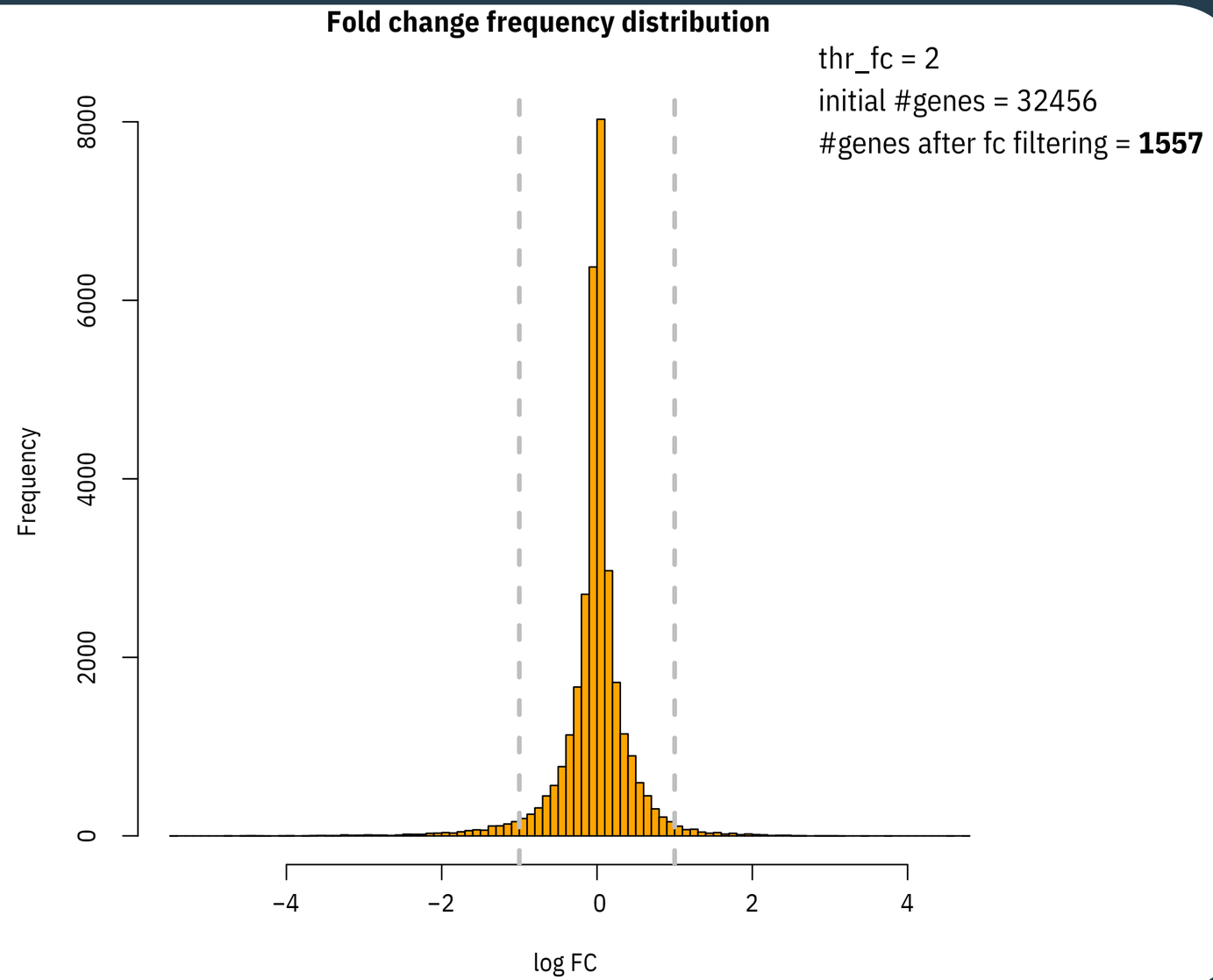


FOLD-CHANGE FILTERING

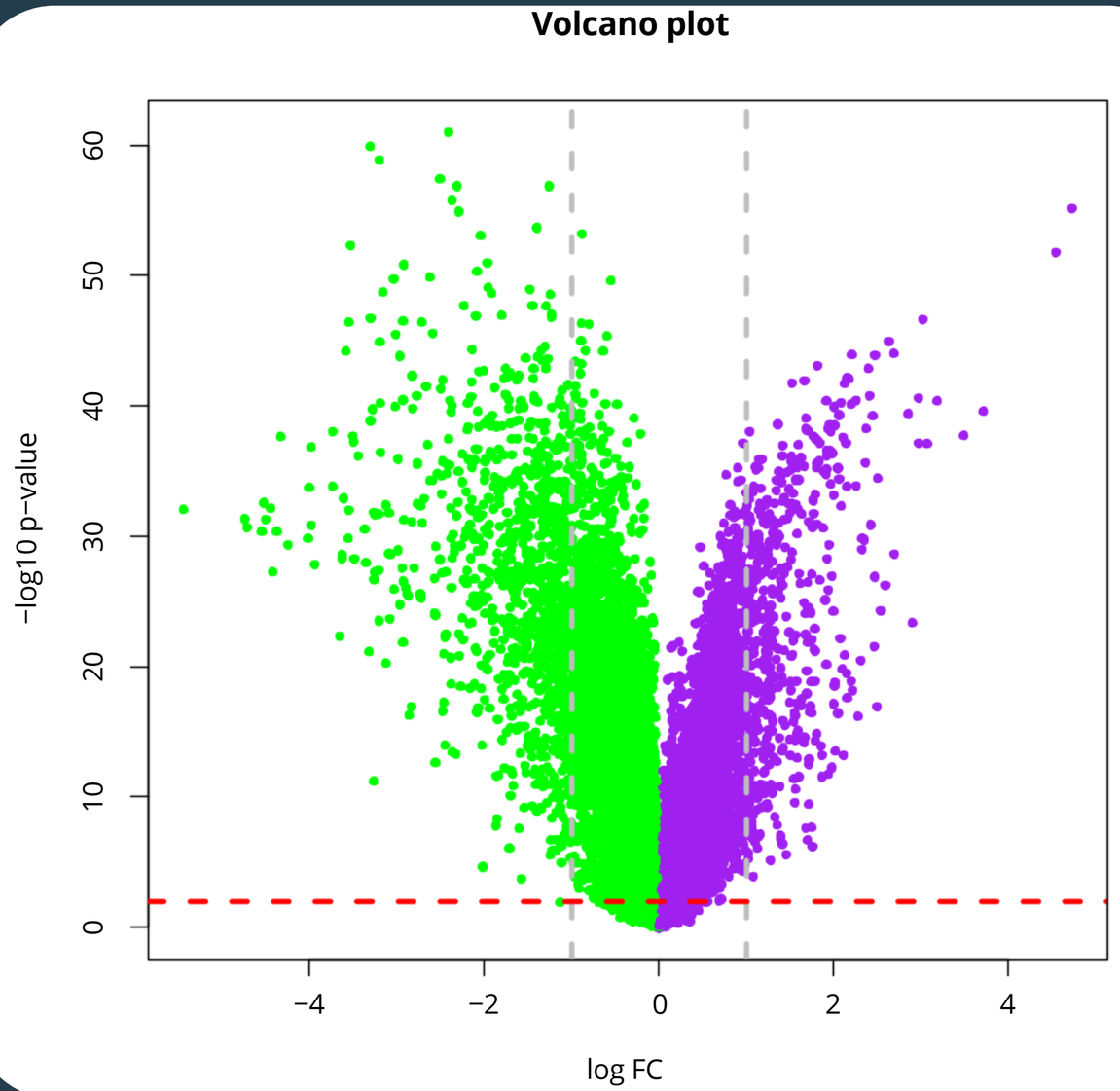
1. Compute fold change as:
   logFC = log(mean(case)) - log(mean(control))
2. Remove all genes such that |logFC| < log(thr_fc)
3. Compute p-values with paired t-test
   (#samples = 112 > 30 so we can assume normal distribution)
4. Adjust p-values with False Discovery Rate method
5. Remove genes with adjusted-pvalue greater than the significance level thr_pval = 0.01



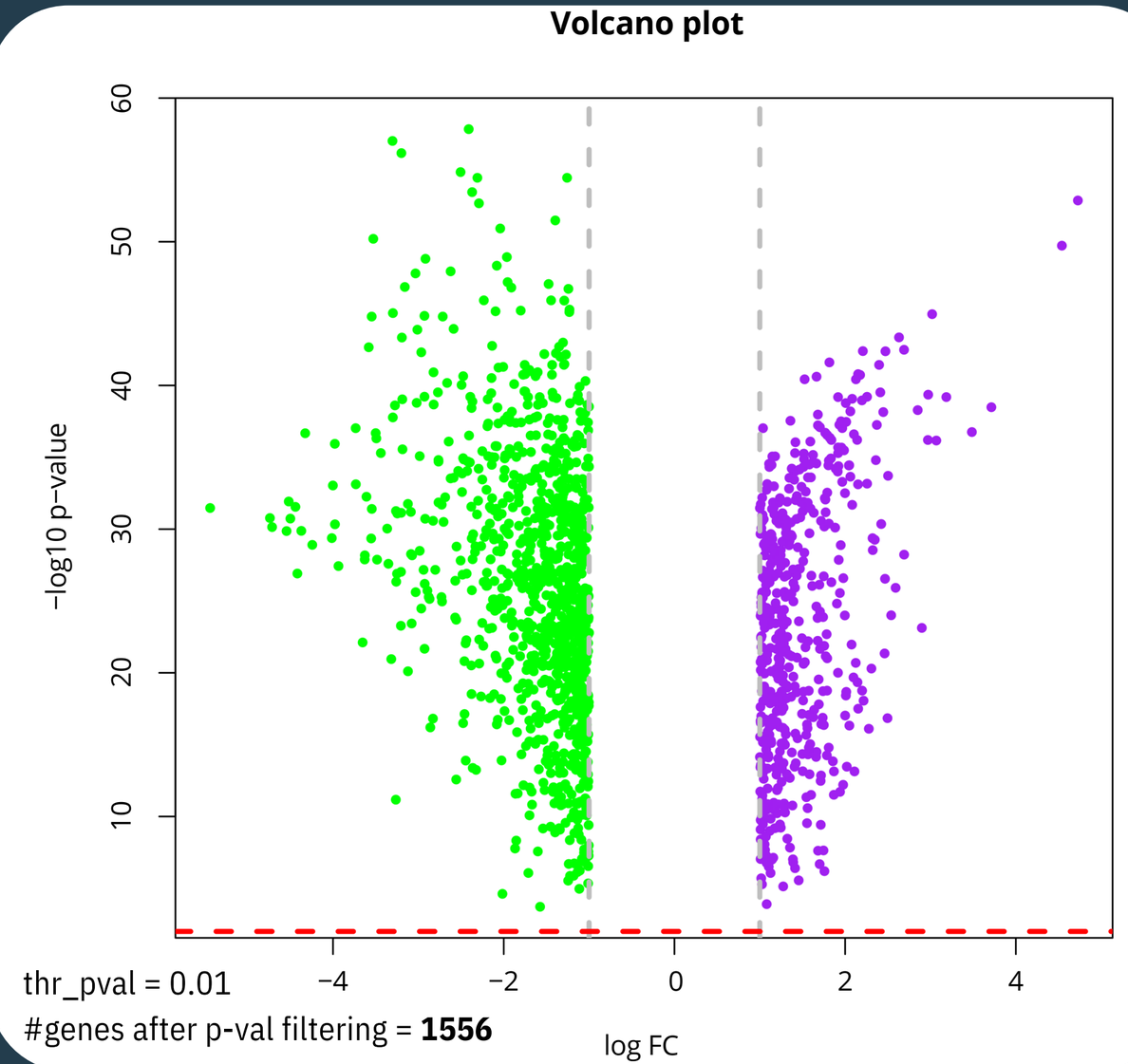
FOLD-CHANGE


Fold change frequency distribution
thr_fc = 2
initial #genes = 32456
#genes after fc filtering = 1557
Frequency
0
2000
4000
6000
8000
−4
−2
0
2
4
log FC

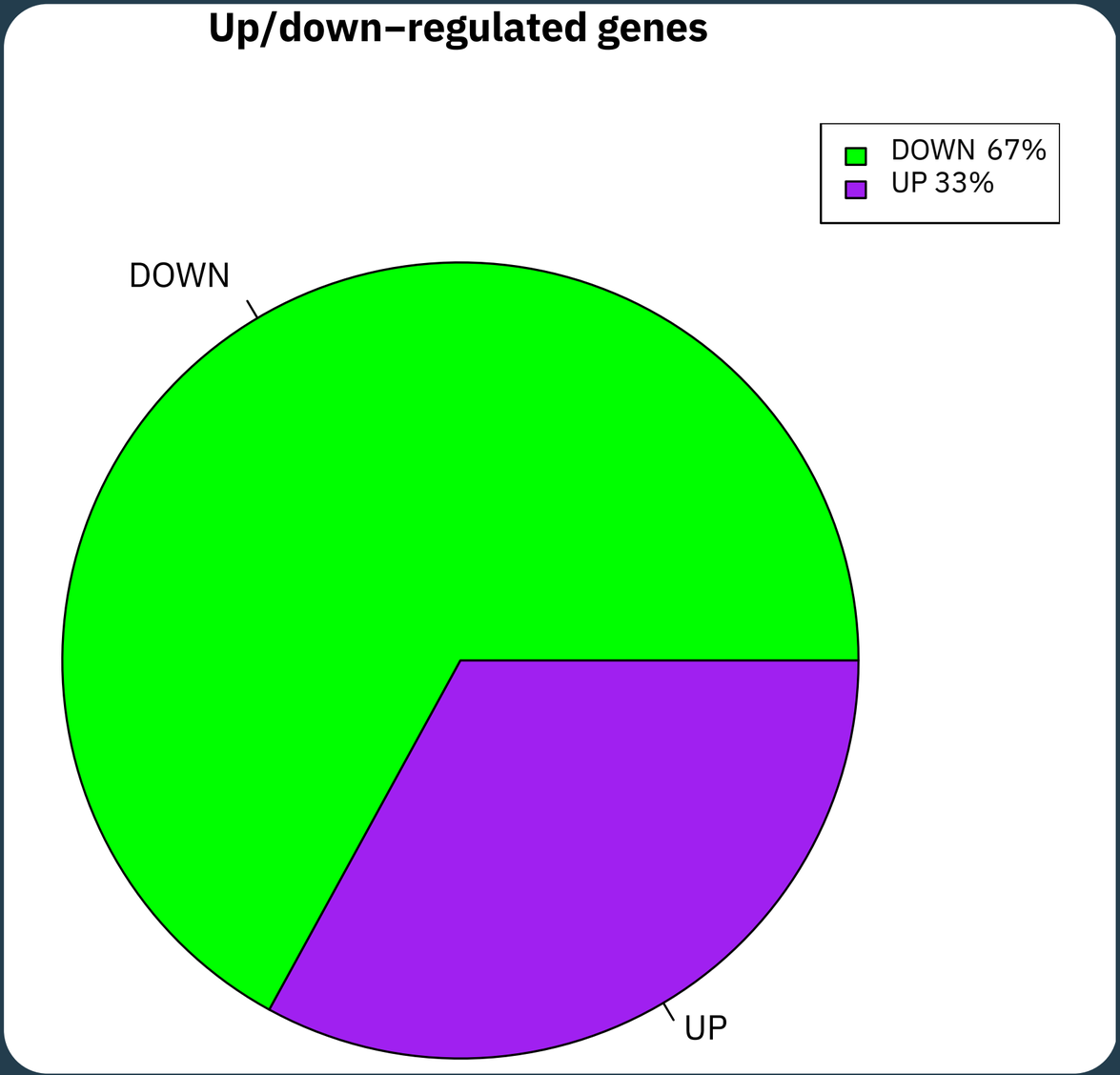
FC Volcano plot



p-value filtering

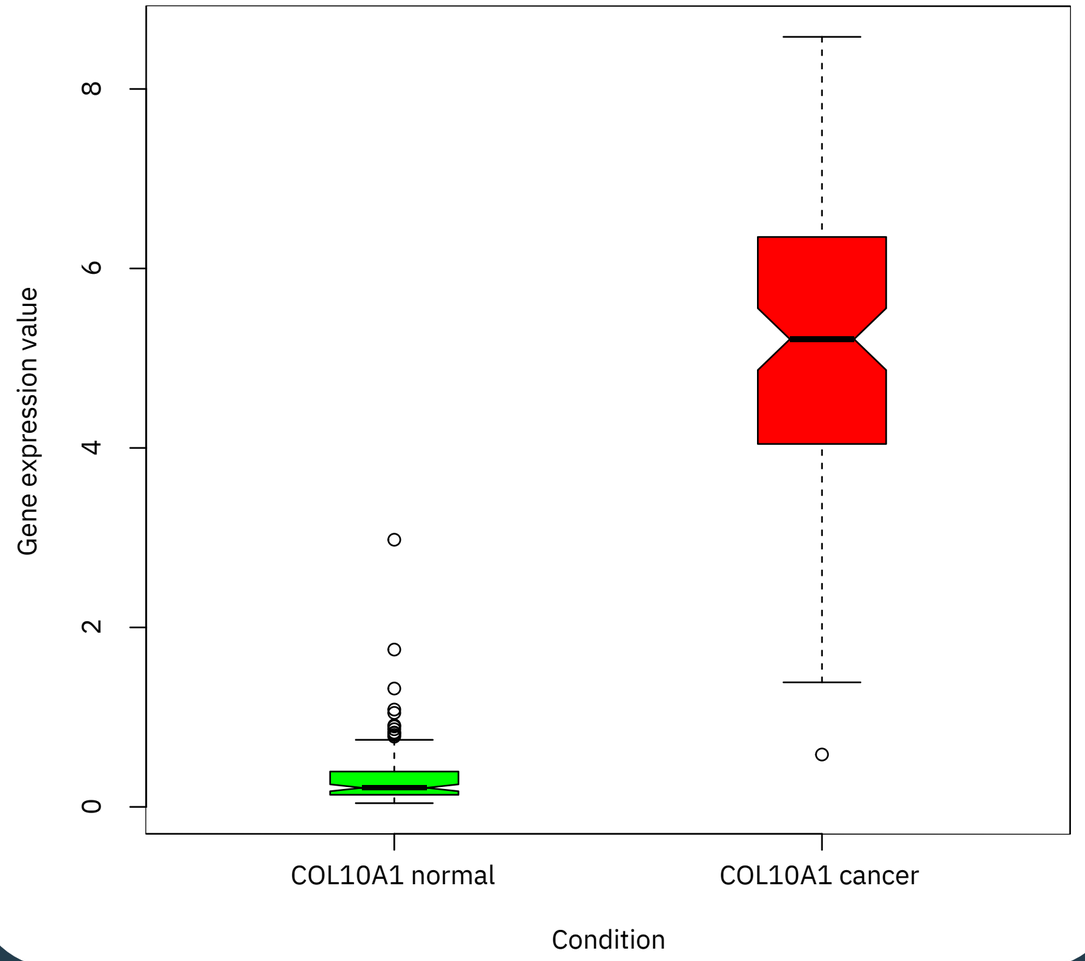


UP/DOWN REGULATED GENES: PIE CHART

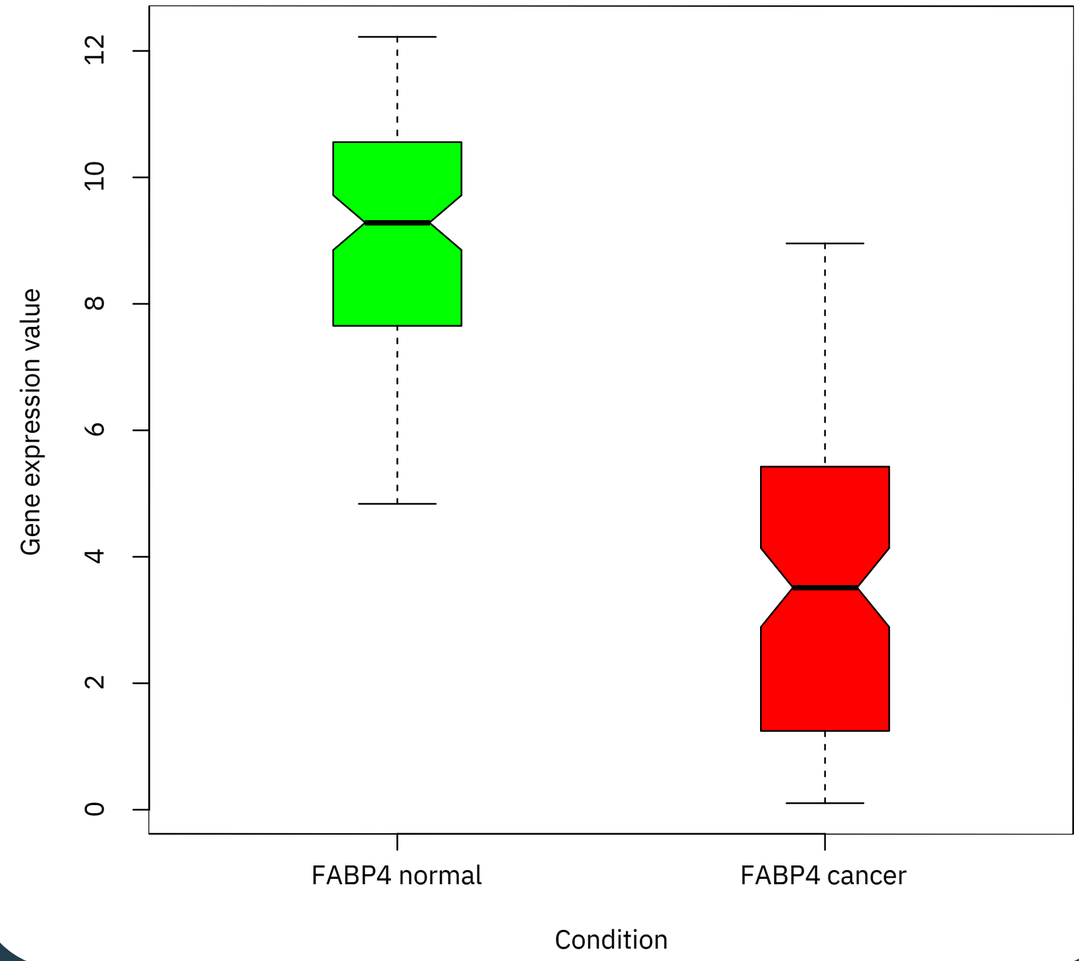


The pie chart shows the ratio between up and down regulated genes, after the filtering process

MOST DEREGULATED GENES

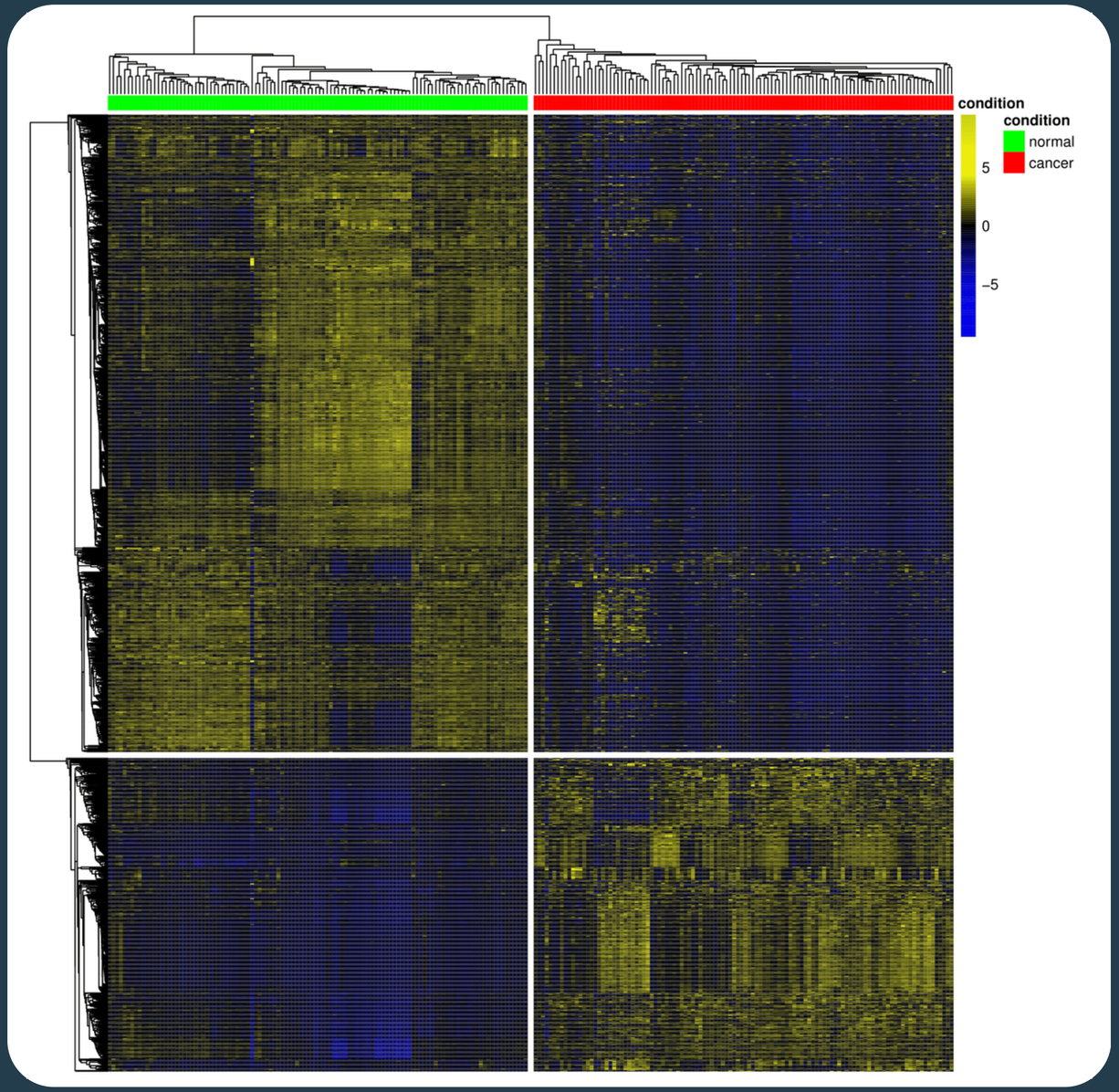


Most Up-regulated gene **COL10A1**



Most Down-regulated gene **FABP4**

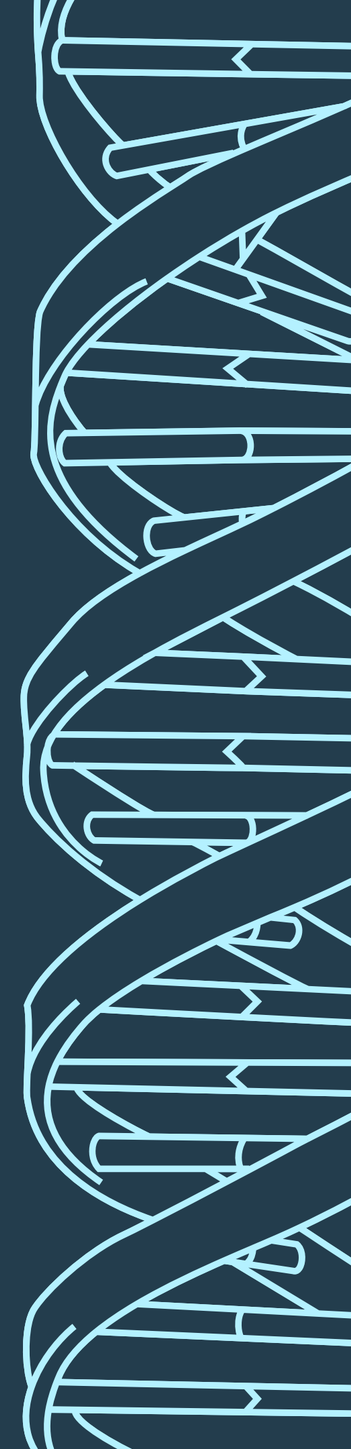
HEAT MAP



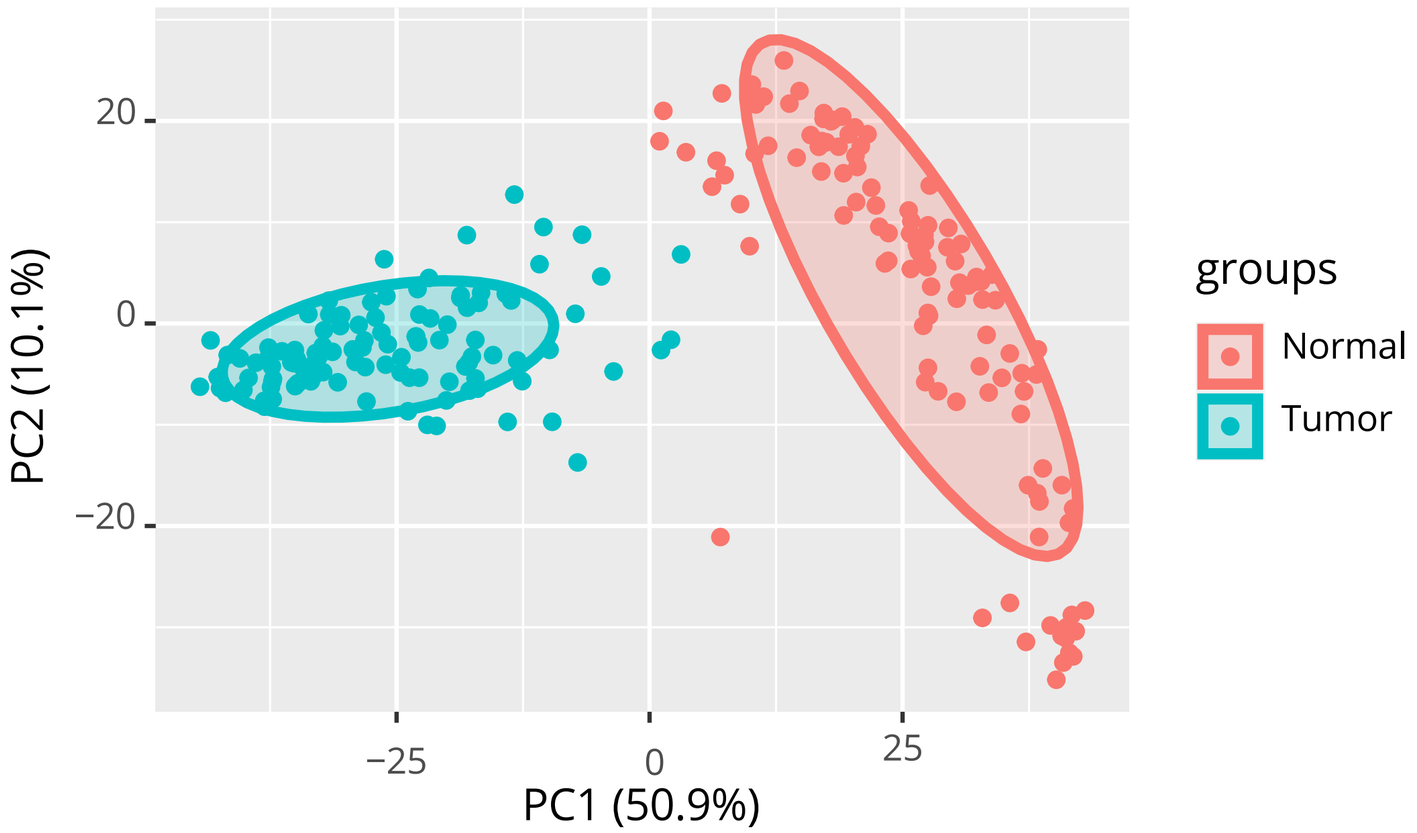
The heat map shows a perfect division between normal and cancer samples.

You can also see the map is clearly divided into 4 regions: a subset of underexpressed genes, and another subset of overexpressed genes in cancer samples with rerspect to the normal ones

PRINCIPAL COMPONENTS ANALYSIS



SCORE PLOT

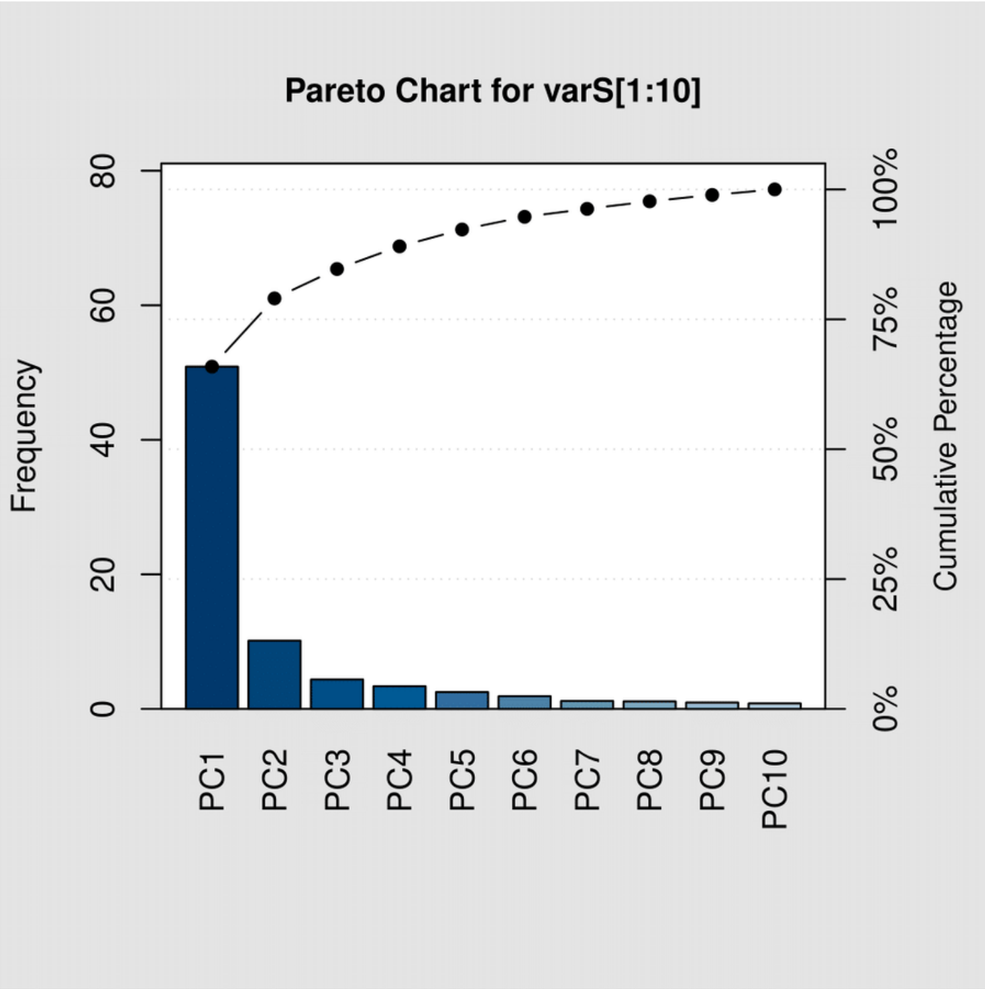


- The score plot shows two principal components, PC1 and PC2, which are the axes of the graph. PC1 accounts for 50.9% of the variance in the dataset, while PC2 accounts for 10.1%.

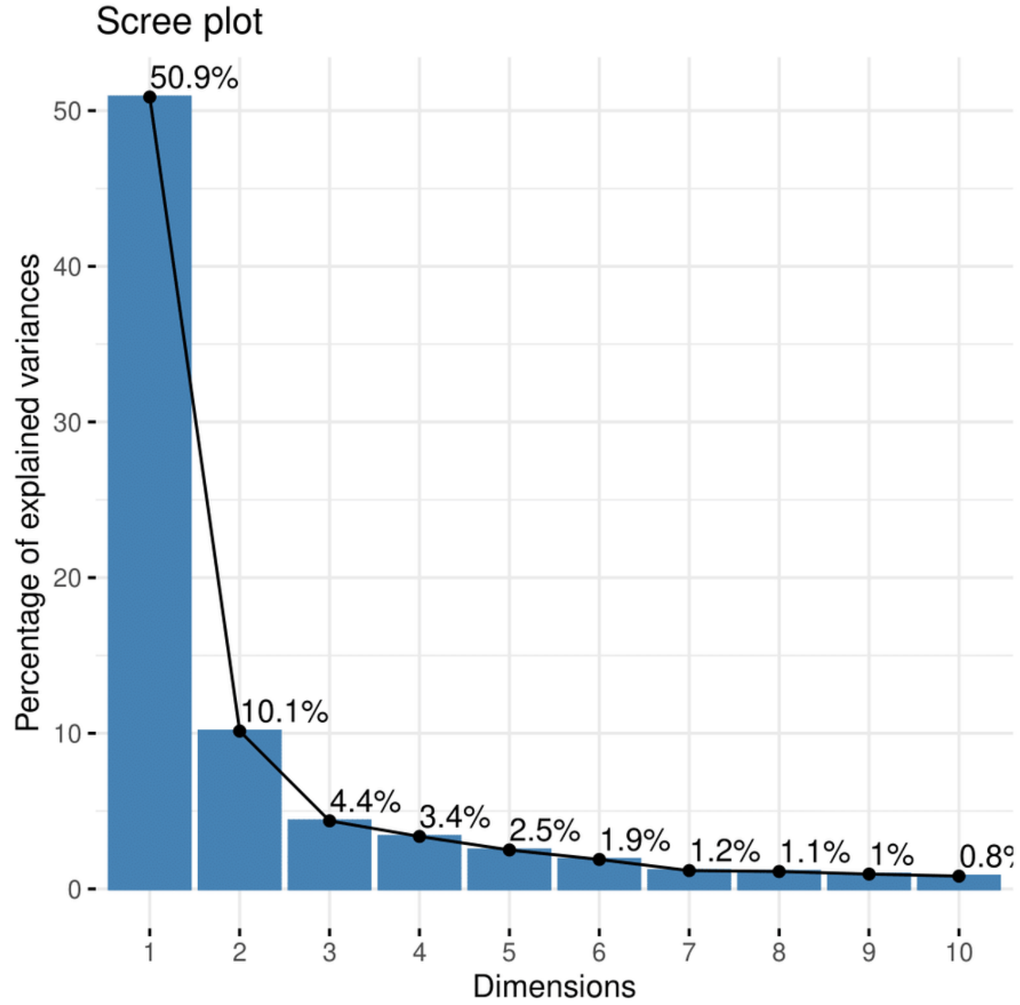
- The plot shows a clear separation between the two groups, indicating that their gene expression profiles are distinct enough to be differentiated by PCA.

- The ellipses represent the spread and clustering of the data points within each group. The 'Tumor' group appears to be more tightly clustered than the 'Normal' group, which is more spread out. This could suggest greater variability in gene expression profiles among normal samples compared to tumor samples.

PRINCIPAL COMPONENTS VARIABILITY

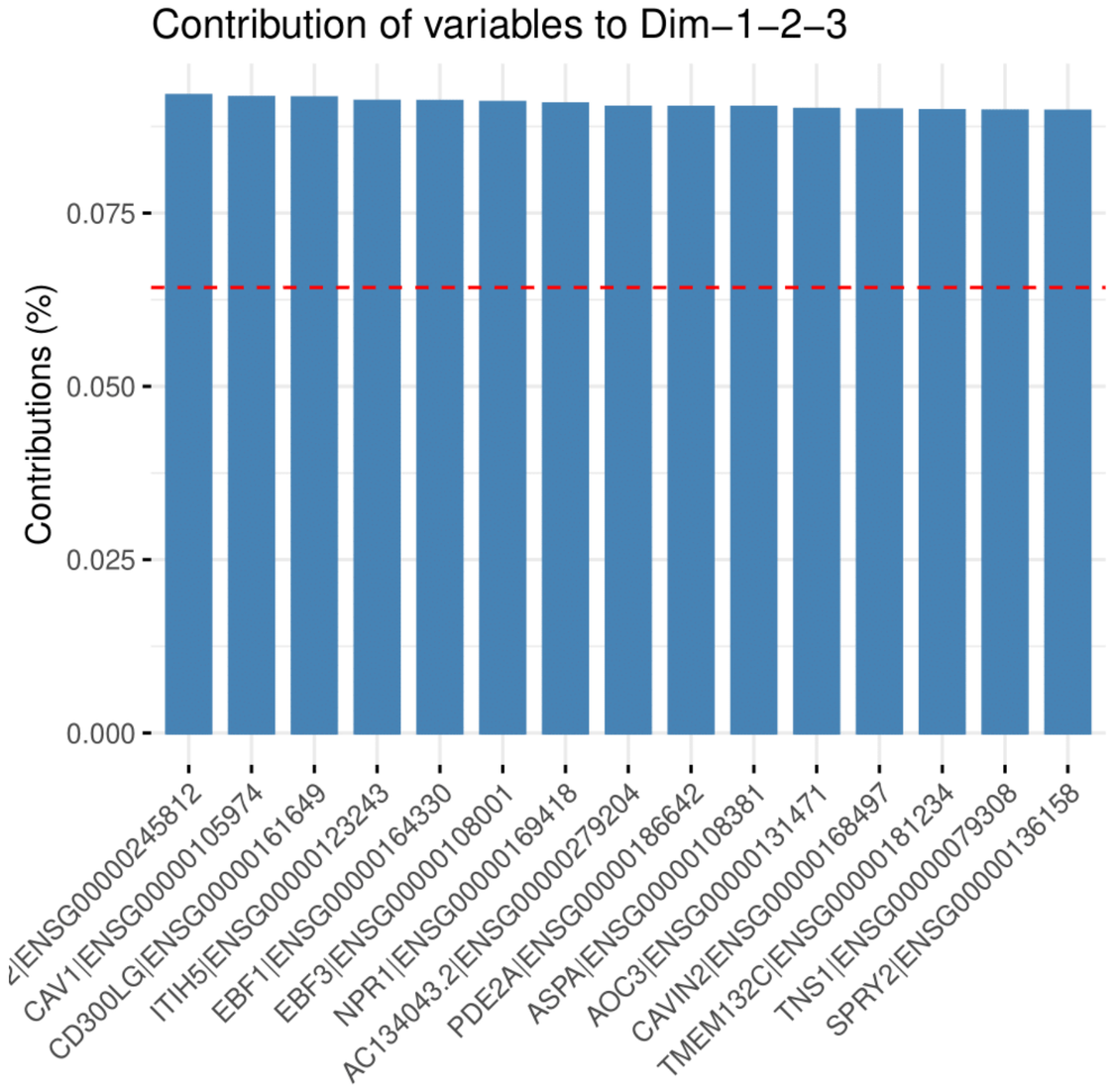


The cumulative percentage (line graph) reaches 75% by PC3 or PC4, suggesting that most of the variability in the data can be captured by the first few components



From this plot it is easy to find the "elbow" point, suggesting that subsequent components have a low contribution to the variance and may be ignored. In this case, the elbow appears to occur after the third component

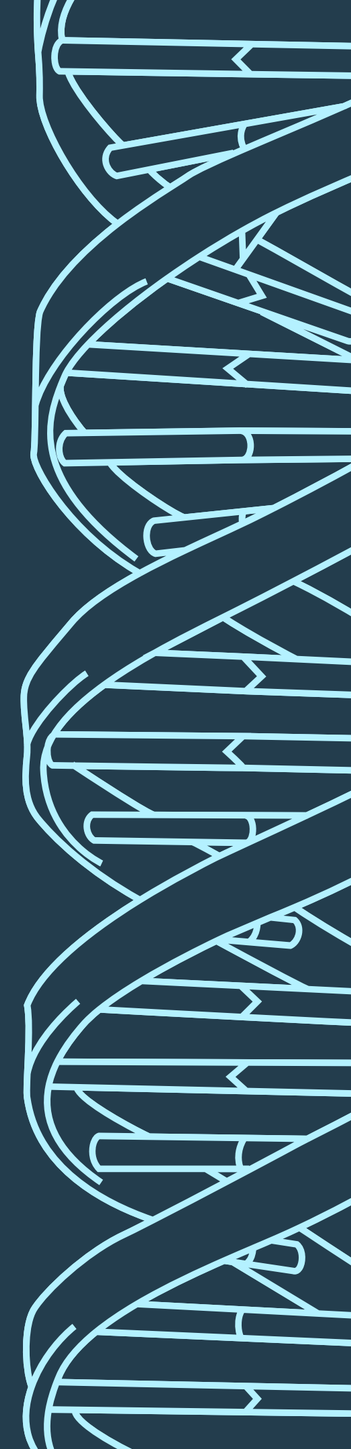
COMPLONENTS CONTRIBUTION



Each bar corresponds to a different gene, the height of each bar represents the percentage contribution of that particular variable to the selected dimensions (Dimension 1, 2, and 3).

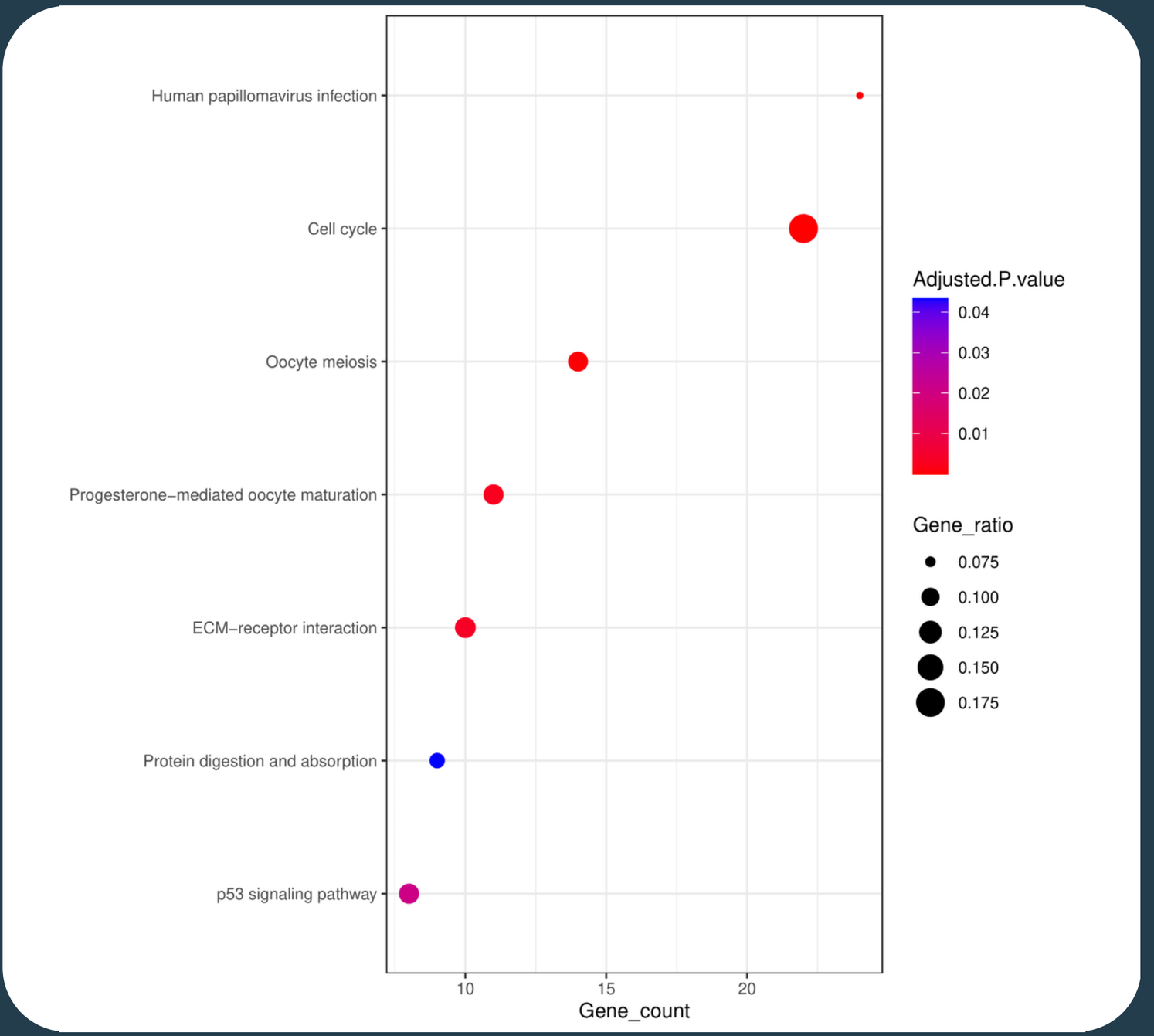
From this analysis we can see that **LINC02202** and **CAV1** are the two genes that contribute the most.

FUNCTIONAL ENRICHMENT ANALYSIS



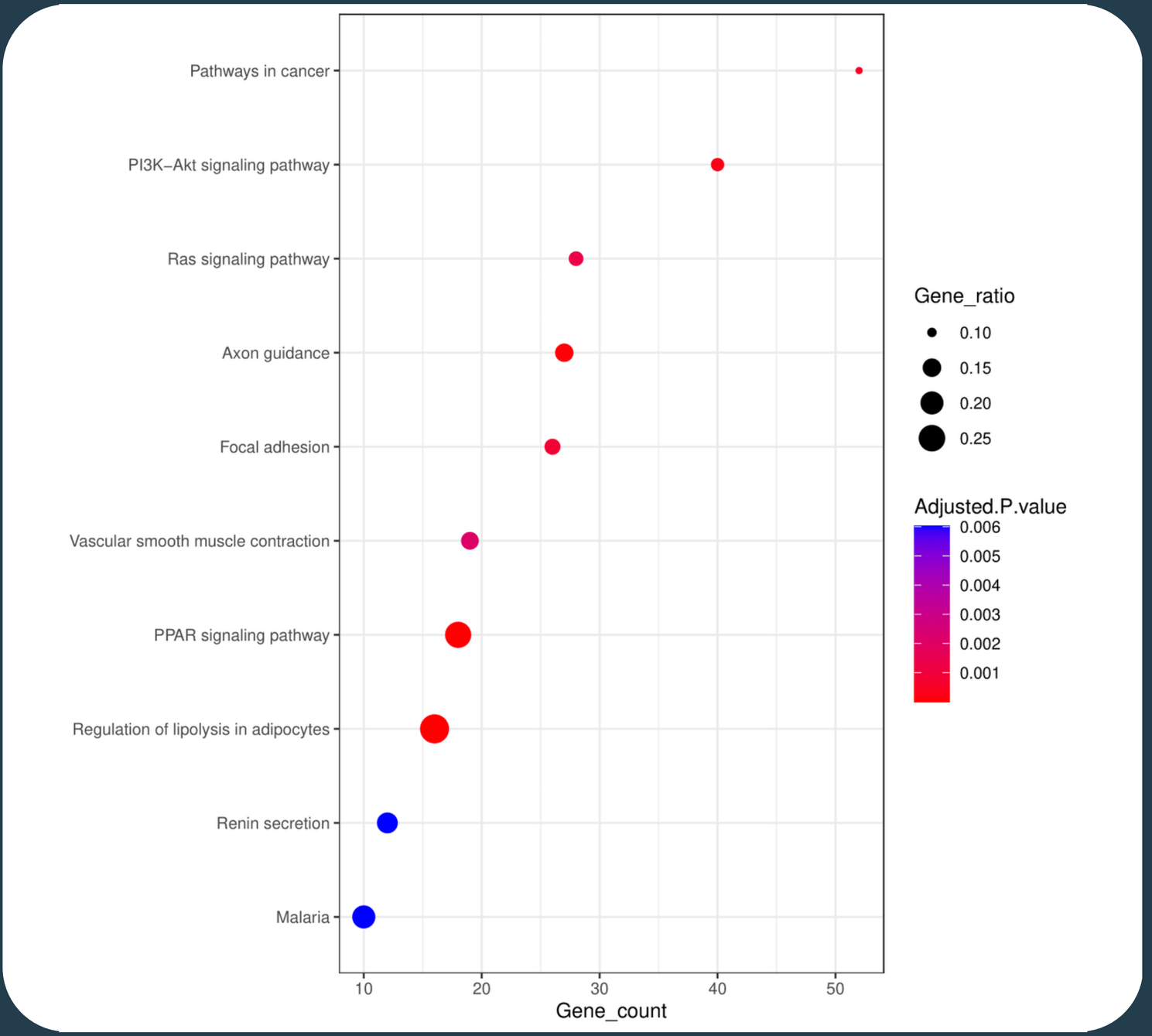
KEGG 2021 HUMAN

The query result shows that **Cell Cycle** is definitely the most enriched pathway by the **up-regulated genes,** with respect to the adjusted p-value, followed by **Human papilloma infection**



KEGG 2021 HUMAN

The query result shows that **Regulation of lipolysis in adipocytes**, **PPAR signaling pathway** and **Axon guidance** are the most enriched pathways by the **down-regulated genes**, with respect to the adjusted p-value

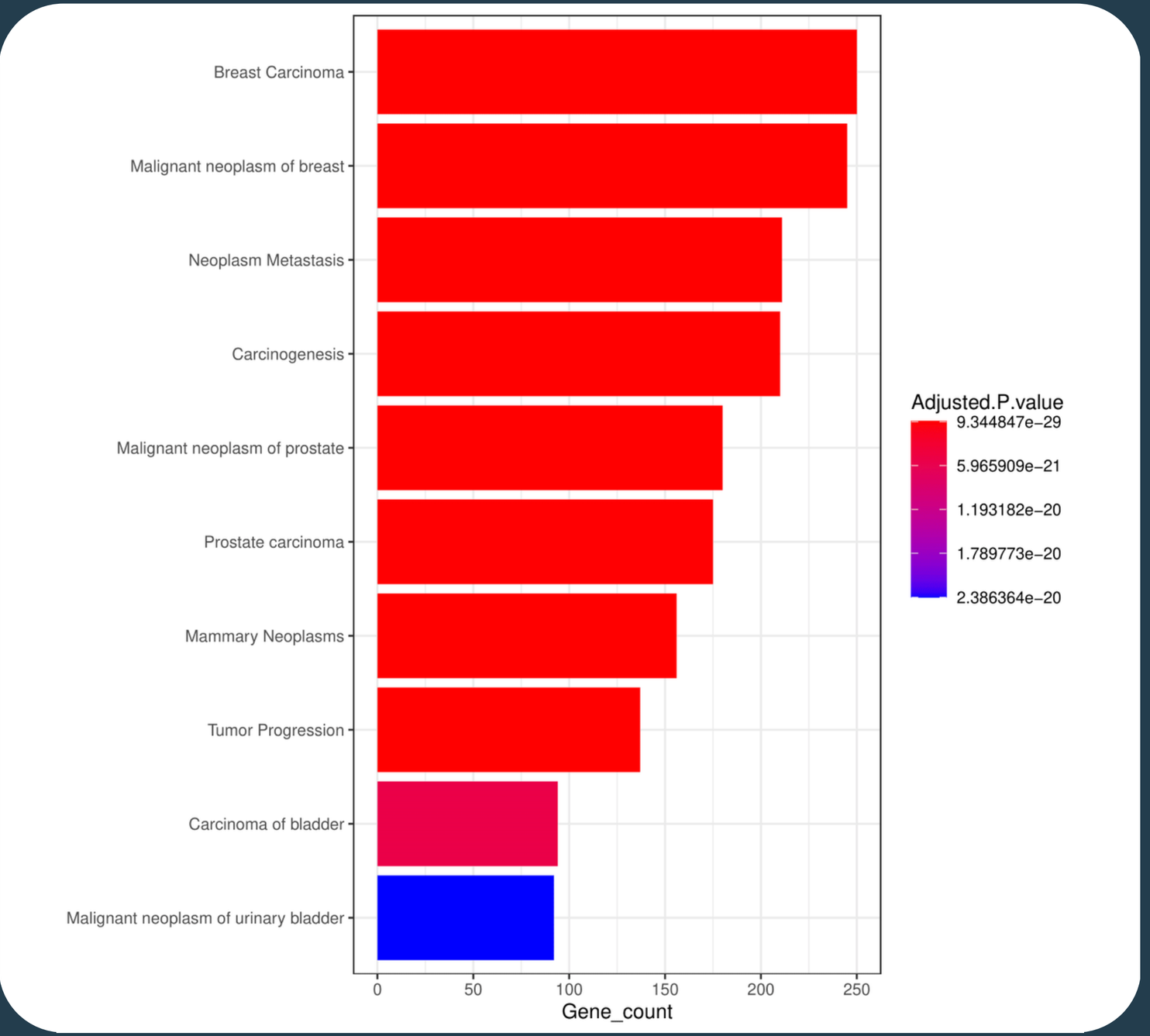


DISGENET: UP-REGULATED

The results of this query have a strong correlation with breast invasive carcinoma.

The most relevant by adjusted p-value are:

- **Breast Carcinoma**
- **Neoplasm Metastasis**
- **Mammary Neoplasms**
- **Malignant neoplasm of breast**
- **Carcinogenesis**

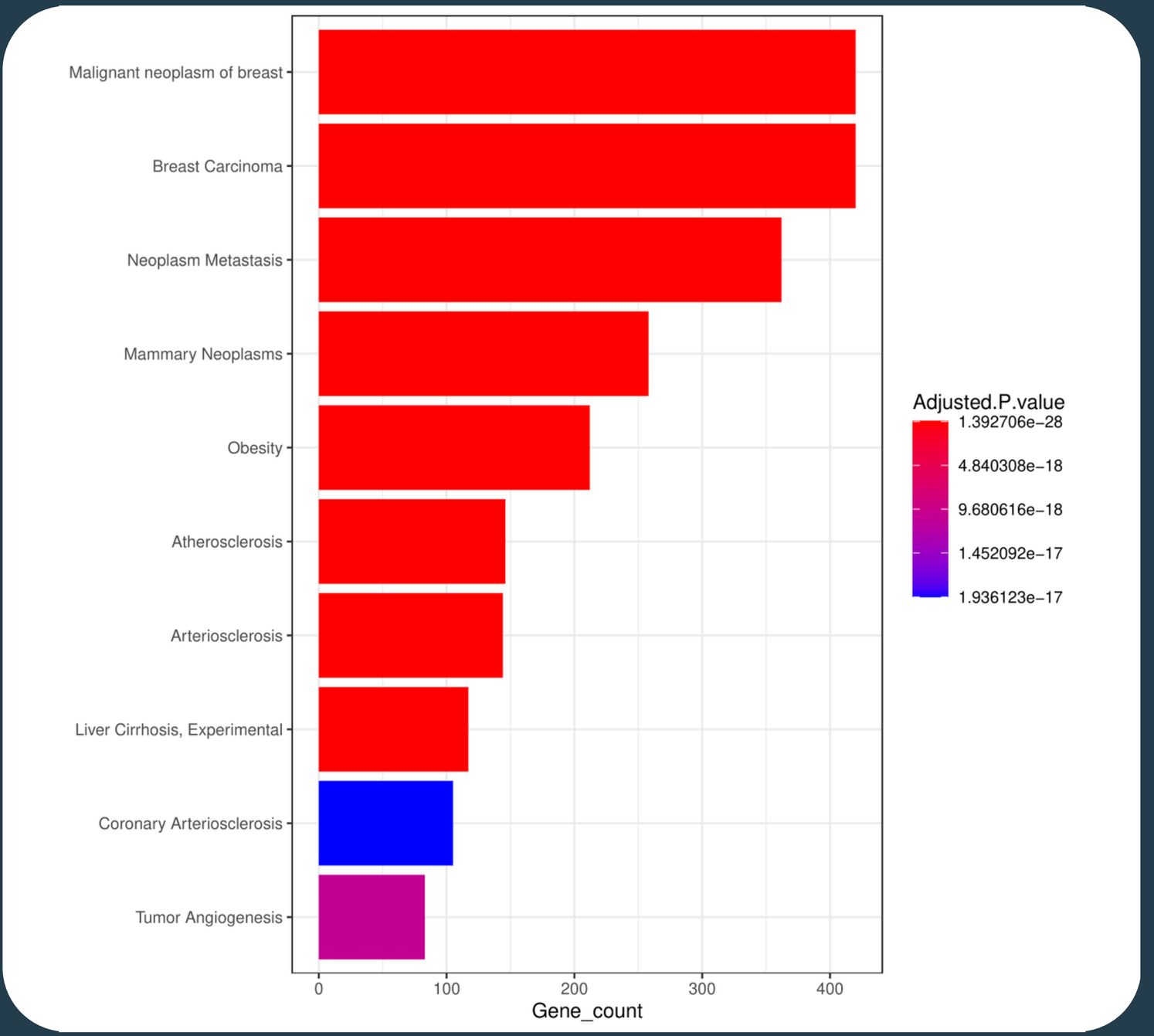


DISGENET: DOWN-REGULATED

The results of this query are very similar to the previous one, underlying a strong correlation with breast invasive carcinoma.

The most relevant by adjusted p-value are:

- **Neoplasm Metastasis**
- **Mammary Neoplasms**
- **Breast Carcinoma**
- **Malignant neoplasm of breast**



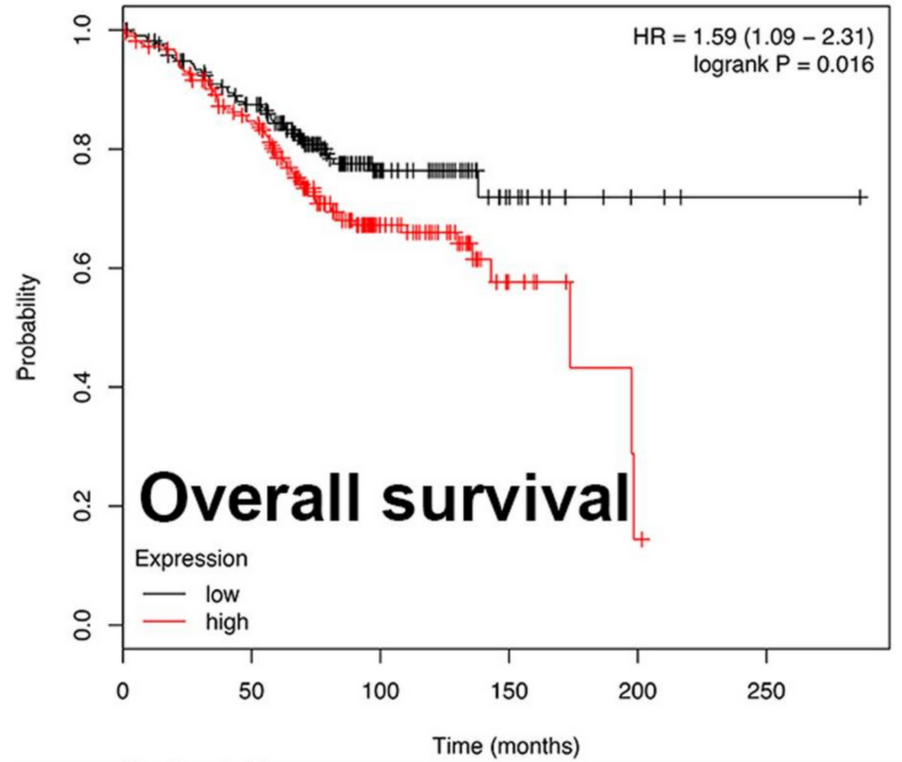
RESULTS ANALYSIS

THE MOST UP-REGULATED GENE: COL10A1

This gene encodes the **alpha chain of type X collagen**, a short chain collagen expressed by hypertrophic chondrocytes during endochondral ossification.

Endochondral ossification is the process by which growing cartilage is systematically replaced by bone to form the growing skeleton.

- Collagen type X alpha 1 (COL10A1) is **overexpressed** in diverse tumors and displays vital roles in **tumorigenesis**.
- COL10A1 protein levels in plasma might be a potential **diagnostic predictor** for early breast cancer.
- Patients with increased COL10A1 expression level showed **worse overall survival** (OS).



Zhang M, Chen H, Wang M, Bai F, Wu K. Bioinformatics analysis of prognostic significance of COL10A1 in breast cancer. Biosci Rep. 2020 Feb 28;40(2):BSR20193286. doi: 10.1042/BSR20193286. PMID: 32043519; PMCID: PMC7029149.

THE MOST DOWN-REGULATED GENE: FABP4

Cancer cells **increase lipolysis of adipocytes** which secrete FABPs through extracellular vesicles. **Cancer cells use FAs generated from lipolysis as fuel to increase growth and invasiveness**.

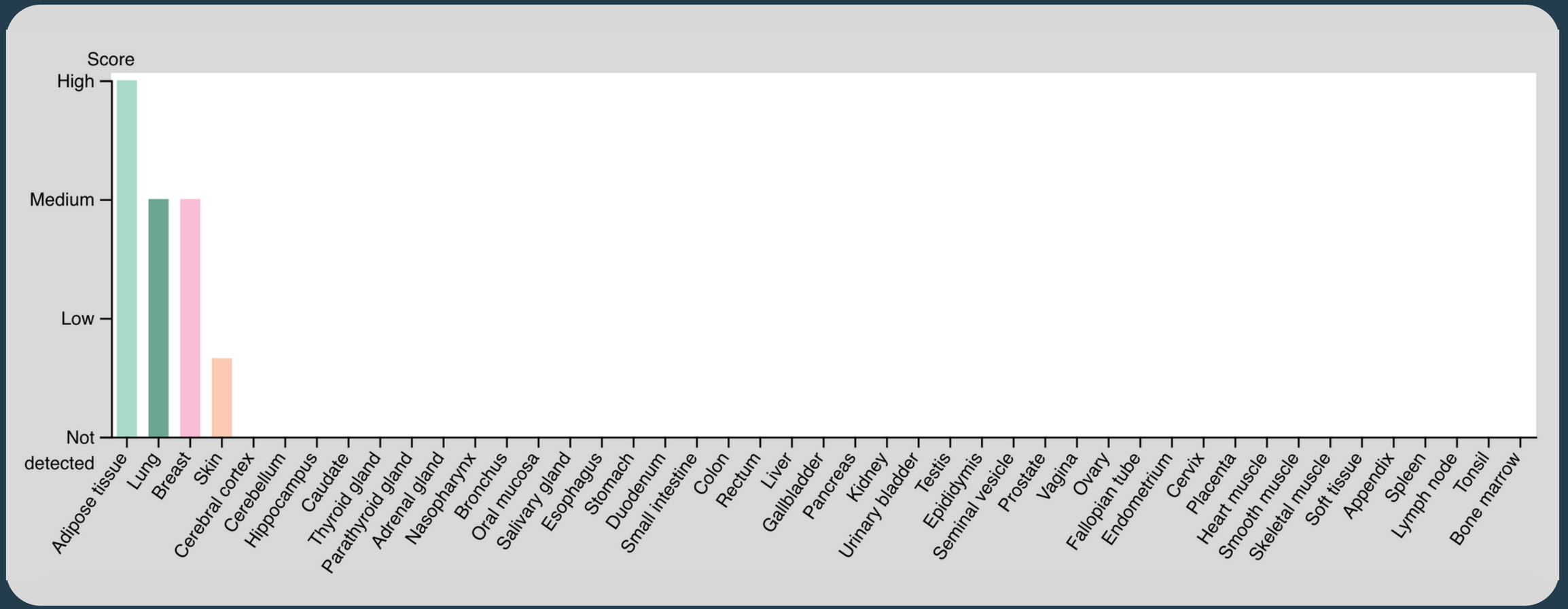
The exact mechanisms of communication between adipocytes and tumor cells are not completely understood. FABPs have been shown to play a role in **transporting FAs between adipocytes and cancer cells**.

As a result, cancer cells **upregulate synthesis of FAs**, and elevated levels of fatty acid synthase (FAS) correlate negatively with cancer prognosis. Besides synthesizing FAs, it has become clear that **cancer cells can also use exogenous FAs for fuel as well**.

So FABP4 is typically overexpressed in tumor samples. **The mismatch of expression in the dataset may be caused by samples consisting mostly of epithelial tumor cells with lower adipocyte content**, leading to lower overall FABP4 expression in tumor samples compared to normal tissue.

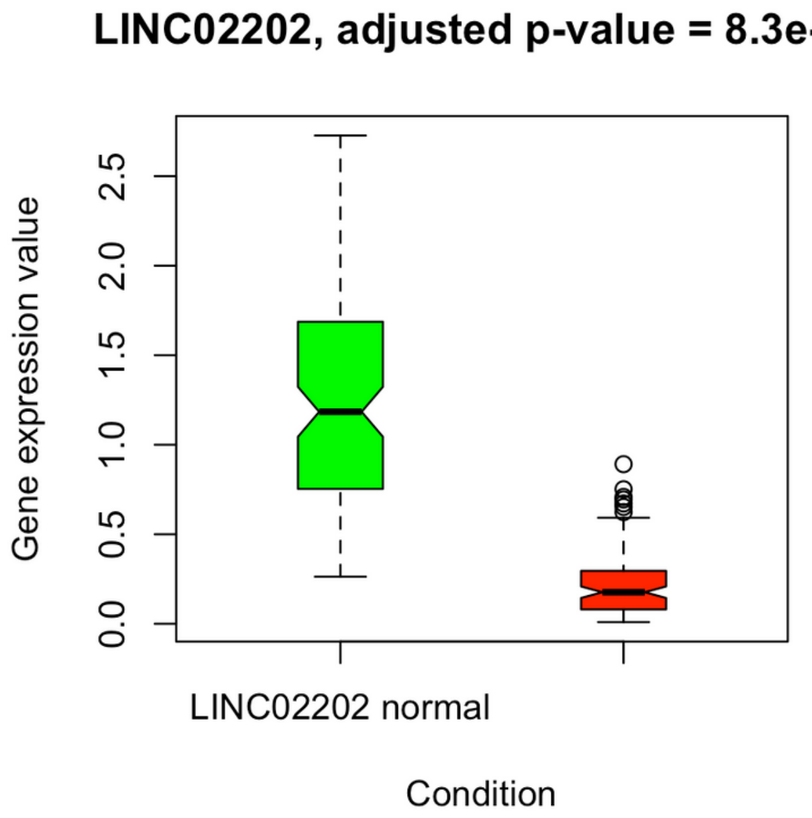
Yang A, Mottillo EP. Adipocyte lipolysis: from molecular mechanisms of regulation to disease and therapeutics. Biochem J. 2020 Mar 13;477(5):985-1008. doi: 10.1042/BCJ20190468. PMID: 32168372; PMCID: PMC7187988.

FABP4: PROTEIN EXPRESSION



LINC02202

Long intergenic non-protein coding RNA 2202 is a **long non coding RNA** sequence that is more present in **adipose tissue**, and it is strongly correlated with **breast cancer**. However, even if the specific biological processes are still not clear, LINC02202 could be a **potential diagnostic biomarker**.



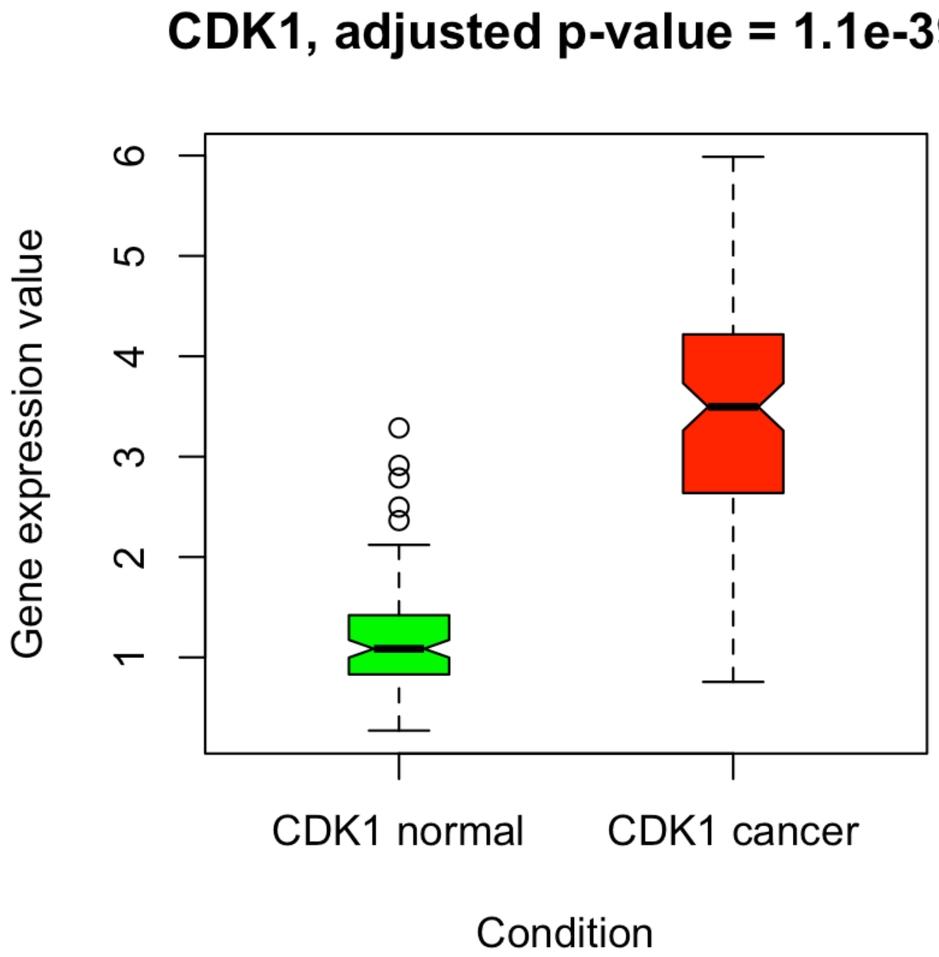
CAV1

The **scaffolding protein** encoded by this gene is the main component of the caveolae plasma membranes found in most cell types. The protein links integrin subunits to the tyrosine kinase FYN, promoting cell cycle progression. Caveolin 1 and caveolin 2 are located next to each other on chromosome 7 and express colocalizing proteins that form a stable hetero-oligomeric complex.

CAV1 can act as a **suppressor** in breast cancer cell carcinogenic process via suppressing breast cancer cell proliferation, autophagy, invasion and migration and promoting apoptosis. CAV1 can also act as a **promoter** in breast cancer cell carcinogenic process via promoting breast cancer cell proliferation, autophagy, invasion, migration and metastasis, and suppressing apoptosis and anoikis.

Qian XL, Pan YH, Huang QY, Shi YB, Huang QY, Hu ZZ, Xiong LX. Caveolin-1: a multifaceted driver of breast cancer progression and its application in clinical treatment. Onco Targets Ther. 2019 Feb 27;12:1539-1552. doi: 10.2147/OTT.S191317. PMID: 30881011; PMCID: PMC6398418.

CDK1



The protein encoded by this gene is a catalytic subunit of the highly conserved protein kinase complex known as M-phase promoting factor (MPF), which is **essential for G2/M phase transitions of eukaryotic cell cycle**. Mitotic cyclins stably associate with this protein and function as regulatory subunits. The phosphorylation and dephosphorylation of this protein also play important regulatory roles in cell cycle control.

CDK1 N THE CELL CYCLE PATHWAY

Role in the Cell Cycle:

CDK1 is a key player in cell cycle regulation. It forms a complex with cyclin B, driving the cell from the G2 phase into mitosis. This transition is critical for cell division.

Upregulation in Cancer:

The overexpression of CDK1 is frequently observed in various cancers. This can lead to unchecked cell proliferation. The aberrant activity of CDK1 can cause genomic instability and resistance to apoptosis.

Therapeutic Target:

Given its central role in cell cycle progression, CDK1 is a potential target for cancer therapy. Inhibitors of CDK1 can potentially halt the proliferation of cancer cells.

CDK1 IN THE P53 SIGNALING PATHWAY

Interaction with P53 Pathway:

The P53 protein is a well-known tumor suppressor. It can induce cell cycle arrest, DNA repair, and apoptosis in response to cellular stress and DNA damage.

CDK1 and p53 Dynamics:

In some contexts, CDK1 can phosphorylate and inhibit proteins that are involved in the p53 pathway, thereby impacting the cell's response to DNA damage and its ability to undergo apoptosis.

Downregulation in the Context of p53:

Although CDK1 is upregulated in cancer, its activity can be modulated by the p53 pathway. The presence of functional p53 can suppress the activity of CDK1, leading to cell cycle arrest and apoptosis. Conversely, in many cancers, the p53 pathway is often mutated or inactivated, leading to the unchecked activity of CDK1.

IMPLICATIONS IN CANCER

The dual role of CDK1 in promoting cell cycle progression and interacting with the P53 signaling pathway, especially in the context of its upregulation in cancer, makes it a critical factor in tumorigenesis. This dual involvement presents a complex scenario where CDK1 promotes cell proliferation while also potentially impacting the cell's ability to undergo apoptosis in response to DNA damage.

From the dataset results there is no statistical evidence of any difference in the expression of TP53 between cancer and normal samples.

